

Financial Statement Analysis

Examples of External Uses of Statement Analysis

- The firm itself and outside providers of capital – creditors and investors – all undertake financial statement analysis. The type of analysis varies according to the specific interests of the party involved.
- Trade creditors (suppliers owed money for goods and services) are primarily interested in the liquidity of a firm. Their claims are short term, and the ability of the firm to pay these claims quickly is best judged by an analysis of the firm's liquidity.
- The claims of bondholders, on the other hand, are long term. Accordingly, bondholders are more interested in the cash-flow ability of the firm to service debt over a long period of time. They may evaluate this ability by analyzing the capital structure of the firm, the major sources and uses of funds, the firm's profitability over time, and projections of future profitability.
- Investors in a company's common stock are principally concerned with present and expected future earnings as well as with the stability of these earnings about a trend line. As a result, investors usually focus on analyzing profitability. They would also be concerned with the firm's financial condition insofar as it affects the ability of the firm to pay dividends and avoid bankruptcy.

Internal Uses of Statement Analysis

- Internally, management also employs financial analysis for the purpose of internal control and to better provide what capital suppliers seek in financial condition and performance from the firm.
- From an internal control standpoint, management needs to undertake financial analysis in order to plan and control effectively.
- To plan for the future, the financial manager must assess the firm's present financial position and evaluate opportunities in relation to this current position.
- With respect to internal control, the financial manager is particularly concerned with the return on investment provided by the various assets of the company, and with the efficiency of asset management.
- To bargain effectively for outside funds, the financial manager needs to be attuned to all aspects of financial analysis that outside suppliers of capital use in evaluating the firm

Global Accounting Standards

- Convergence of Accounting Standards: Aims to narrow or remove differences so that investors can better understand financial statements prepared under different accounting frameworks
- IASB – International Accounting Standards Board has the responsibility of IFRS
- IFRS – International Financial Reporting Standards (EU countries adopted)
- US GAAP – US Generally Accepted Accounting Principles determined by FASB
- FASB – Financial Accounting Standards Board determines accounting standards for financial statements

Financial Statements

- Financial (statement) analysis The art of transforming data from financial statements into information that is useful for informed decision making.
- The four key financial statements are
 - (1) the income statement,
 - (2) the balance sheet,
 - (3) the statement of stockholders' equity,
 - (4) the statement of cash flows.

Financial Statements

- **Income Statement**

- The income statement summarizes the firm's operating results. A summary of a firm's revenues and expenses over a specified period, ending with net income or loss for the period.

- **Balance Sheet**

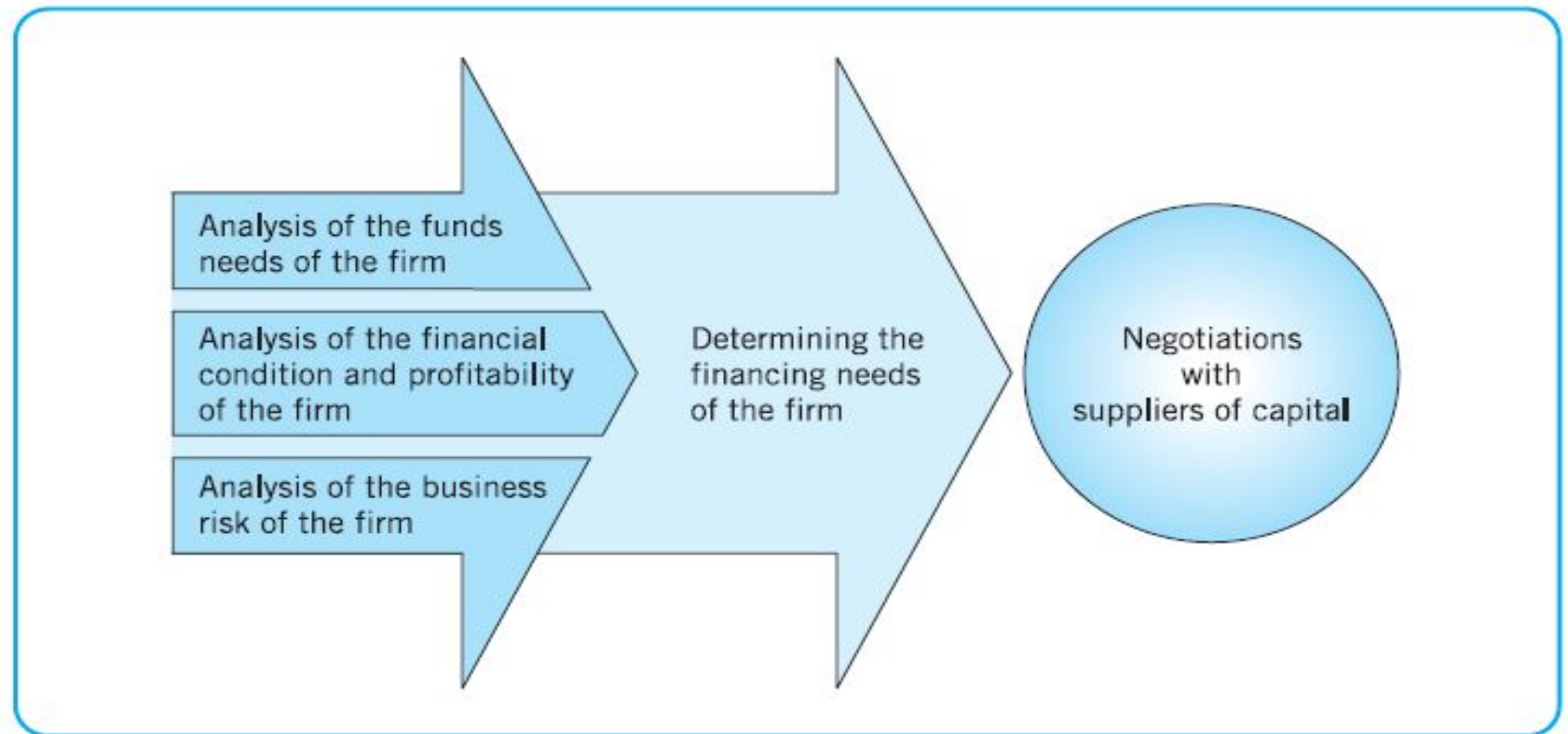
- The balance sheet presents a summary statement of the firm's financial position at a given time. The statement balances the firm's assets (what it owns) against its financing, which can be either debt (what it owes) or equity (what owners provided).

- **Statement of Cash Flows**

- The statement of cash flows is a summary of the cash flows over the period. The statement categorizes cash flows as operating cash flow, investment cash flow, or financing cash flow.

Framework for Financial Analysis

- A number of different approaches might be used in analyzing a firm. Many analysts have a favorite procedure for coming to some generalizations about the firm being analyzed.
- At the risk of treading on some rather sacred ground, we present a conceptual framework that lends itself to situations in which external financing is contemplated.



Framework for Financial Analysis

- Analysis of the funds needs of the firm.
- Our concern in the first case is with the trend and seasonal component of a firm's funds requirements.
- How much funding will be required in the future, and what is the nature of these needs? Is there a seasonal component to the needs?
- Analytical tools used to answer these questions include sources and uses of funds statements, statements of cash flow, and cash budgets,

Framework for Financial Analysis

- Analysis of the financial condition and profitability of the firm.
- The tools used to assess the financial condition and performance of the firm are financial ratios.
- The financial analyst uses these ratios much like a skilled physician uses lab-test results.
- Taken together, and over time, these data offer valuable insights into a firm's health – its financial condition and profitability.
- The Analysis done Individually, Over time, In combination, In comparison

Framework for Financial Analysis

- Analysis of the business risk of the firm.
- Business risk relates to the risk inherent in the operations of the firm.
- Some companies are in highly volatile lines of business and/or may be operating close to their break-even point.
- Other companies are in very stable lines of business and/or find themselves operating far from their break-even point.

Framework for Financial Analysis

- All three of these factors should be used in determining the financial needs of the firm.
- Moreover, they should be considered jointly. The greater the funds requirements, of course, the greater the total financing that will be necessary. The nature of the needs for funds influences the type of financing that should be used.
- If there is a seasonal component to the business, this component lends itself to short-term financing, bank loans in particular. The firm's level of business risk also strongly affects the type of financing that should be used.
- The greater the business risk, the less desirable debt financing usually becomes relative to common stock financing. In other words, equity financing is safer in that there is no contractual obligation to pay interest and principal, as there is with debt

Use of Financial Ratios

- To compare financial results across companies and across time, we need relative performance measures.
- Ratio analysis is a method for evaluating financial performance on a relative basis. The inputs required to conduct ratio analysis come from the firm's income statement and balance sheet.
- Ratio analysis begins with some basic computations, but interpretation of the ratio values is what really matters. Usually, the value of a particular ratio is less important than how it changes over time or how it compares to the same ratios for competing firms.
- Cross-sectional analysis compares different firms' financial ratios at the same point in time. Analysts are often interested in how well a firm has performed relative to other firms in its industry. Nearly all users of financial ratios employ this type of cross-sectional analysis, called **benchmarking**.
- Time-series analysis evaluates performance over time. Comparison of current to past performance, using ratios, enables analysts to assess the firm's progress and to spot trends. Any significant year-to-year changes may indicate a problem, especially if the same trend is not an industry-wide phenomenon.

Cautions About Using Ratio Analysis

- Before discussing specific ratios, we should consider the following cautions about their use:
 1. Ratios that reveal large deviations from the norm merely indicate the possibility of a problem. Analysts must do additional work to determine whether a problem exists and to isolate the causes of the problem.
 2. A single ratio does not generally provide sufficient information on which to judge the overall performance of the firm. However, if an analysis focuses only on certain specific aspects of a firm's financial position, one or two ratios may suffice.
 3. Analysts should use financial statements dated at the same point in the year before calculating ratios for different companies. Otherwise, the effects of seasonality may produce erroneous conclusions.

Cautions About Using Ratio Analysis

4. It is preferable to use audited financial statements for ratio analysis. If statements have not been audited, the data in them may not reflect the firm's true financial condition.
5. The financial data being compared should have been constructed using the same set of accounting principles, such as GAAP. The use of differing accounting treatments can distort the results of ratio comparisons.
6. High inflation can distort results, for example, if the book values of inventory and depreciable assets differ greatly from their replacement values. High inflation tends to make older firms (older assets) appear more efficient and profitable than newer firms (newer assets), so in an inflationary environment comparing older firms to younger ones or even tracking a single firm over a long time can be problematic.

Liquidity Ratios

- The liquidity of a firm reflects its ability to satisfy its short-term obligations as they come due. Generally speaking, a firm with greater liquidity will have an easier time paying its bills and is less likely to become insolvent.
- The two most common liquidity measures are the current ratio and the quick (acid-test) ratio.
- **CURRENT RATIO**
- The **current ratio** measures liquidity by comparing a firm's current assets to its current liabilities. The mathematical expression for the current ratio is

Liquidity Ratios

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$$\text{Current Ratio} = \text{Current Assets} \div \text{Current Liabilities}$$

- QUICK (ACID-TEST) RATIO
- The **quick (acid-test) ratio** is similar to the current ratio except that it excludes inventory, which is generally the least liquid current asset. The generally low liquidity of inventory results from two primary factors

$$\text{Quick Ratio} = (\text{Current Assets} - \text{Inventory}) \div \text{Current Liabilities}$$

Liquidity Ratios

CURRENT RATIO

$$\text{Current Ratio} = \text{Current Assets} \div \text{Current Liabilities}$$

$$2020: \$12,902 \div \$14,487 = 0.89$$

$$2019: \$12,519 \div \$15,014 = 0.83$$

$$2018: \$12,540 \div \$13,052 = 0.96$$

• QUICK (ACID-TEST) RATIO

$$\text{Quick Ratio} = (\text{Current Assets} - \text{Inventory}) \div \text{Current Liabilities}$$

$$2020: (\$12,902 - \$8,992) \div \$14,487 = 0.27$$

$$2019: (\$12,519 - \$9,497) \div \$15,014 = 0.20$$

$$2018: (\$12,540 - \$8,597) \div \$13,052 = 0.30$$

Financial Leverage (Debt) Ratios

- Financial leverage: Refers to the degree to which a firm uses debt financing and to the effects of debt financing.
- The greater the debt, the greater is the firm's risk of being unable to meet its contractual payments. Because firms must satisfy creditors' claims before they can pay shareholders, current and prospective shareholders pay close attention to the firm's ability to repay debts.
- Leverage measures are of two general types: measures of the degree of indebtedness and measures of the ability to repay debts.
- The **degree of indebtedness** ratios measure the amount of debt relative to other significant balance sheet amounts.

Financial Leverage (Debt) Ratios

- The **debt-to-equity ratio** measures the relative proportion of total liabilities to common stock equity used to finance the firm's assets. As with the debt ratio, a higher debt-to-equity ratio means that the firm uses more financial leverage.

$$\text{Debt-to-Equity Ratio} = \text{Total Liabilities} \div \text{Common Stock Equity}$$

$$\frac{\text{Total debt}}{\text{Shareholders' equity}}$$

- The **debt to total Assets ratio** measures the proportion of total assets financed by the firm's creditors. A higher debt ratio means that a firm is using a larger amount of other people's money to finance its operations, or equivalently, the firm is using greater financial leverage.

$$\text{Debt Ratio} = \text{Total Liabilities} \div \text{Total Assets}$$

Financial Leverage (Debt) Ratios

- The **debt-to-equity ratio**

$$\text{Debt-to-Equity Ratio} = \text{Total Liabilities} \div \text{Common Stock Equity}$$

$$2020: \$30,946 \div \$11,833 = 2.62$$

$$2019: \$29,993 \div \$11,297 = 2.65$$

$$2018: \$28,652 \div \$11,651 = 2.46$$

$$\frac{\text{Total debt}}{\text{Shareholders' equity}}$$

- The **debt to total Assets ratio**

$$\text{Debt Ratio} = \text{Total Liabilities} \div \text{Total Assets}$$

$$2020: \$30,946 \div \$42,779 = 0.72$$

$$2019: \$29,993 \div \$41,290 = 0.73$$

$$2018: \$28,652 \div \$40,303 = 0.71$$

Financial Leverage (Debt) Ratios

- We may wish to compute the following ratio, which deals with only the long-term capitalization of the firm. where *total capitalization* represents all long-term debt and shareholders' equity

$$\frac{\text{Long-term debt}}{\text{Total capitalization}}$$

Solvency Ratio (Coverage Ratios)

- *Interest Coverage* measures the firm's ability to make contractual interest payments. The higher its value, the better able the firm is to fulfill its interest obligations.

$$\text{Times Interest Earned Ratio} = \text{Earnings Before Interest and Taxes} \div \text{Interest}$$

$$2020: \quad \$4,658 \div \$477 = 9.77$$

$$2019: \quad \$4,110 \div \$461 = 8.92$$

$$2018: \quad \$4,224 \div \$653 = 6.47$$

Activity or Efficiency Ratios

- **Activity ratios** measure the speed with which various asset and liability accounts convert into sales or cash.
- Activity ratios measure how efficiently a firm manages inventory, disbursements, and collections.
- We can also use activity ratios to assess the efficiency with which a firm manages its total assets.

Activity or Efficiency Ratios

- AVERAGE COLLECTION PERIOD: The **average collection period**, or average age of accounts receivable, is useful in evaluating credit and collection policies. It equals the accounts receivable balance divided by average daily sales

$$\begin{aligned}\text{Average Collection Period} &= \text{Accounts Receivable} \div \text{Average Sales per Day} \\ &= \text{Accounts Receivable} \div (\text{Annual Sales} \div 365) \quad (3.5)\end{aligned}$$

- The *receivable turnover (RT) ratio* provides insight into the quality of the firm's receivables and how successful the firm is in its collections. This ratio is calculated by dividing receivables into annual net credit sales. This ratio tells us the number of times accounts receivable have been turned over (turned into cash) during the year. The higher the turnover, the shorter the time between the typical sale and cash collection.

- *average collection period* = $\frac{\text{Days in the year}}{\text{Receivable turnover}}$

$$\frac{\text{Annual net credit sales}}{\text{Receivables}}$$

Activity or Efficiency Ratios

- The **average collection period**,

$$2020: \$441 \div (\$78,112 \div 365) = 2.06 \text{ days}$$

$$2019: \$428 \div (\$75,356 \div 365) = 2.07 \text{ days}$$

$$2018: \$400 \div (\$72,714 \div 365) = 2.01 \text{ days}$$

Activity or Efficiency Ratios

- Average Payment Period: The average amount of time needed to pay accounts payable.

$$\begin{aligned}\text{Average Payment Period} &= \text{Accounts Payable} \div \text{Average Purchases per Day} \\ &= \text{Accounts Payable} \div (\text{Annual Purchases} \div 365)\end{aligned}\quad (3.6)$$

- *Payable turnover (PT) ratio* (annual credit purchases divided by accounts payable), allows us to analyze payables in much the same manner as we analyze receivables. Also, we can compute the *payable turnover in days (PTD)* or *average payable period* as

$$\frac{\text{Days in the year}}{\text{Payable turnover}}$$

or, equivalently,

$$\frac{\text{Accounts payable} \times \text{Days in the year}}{\text{Annual credit purchases}}$$

Activity or Efficiency Ratios

- Average Payment Period: The average amount of time needed to pay accounts payable.

$$2020: \$9,920 \div [(0.90 \times \$54,864) \div 365] = 73.3 \text{ days}$$

$$2019: \$9,761 \div [(0.90 \times \$53,299) \div 365] = 74.3 \text{ days}$$

$$2018: \$8,677 \div [(0.90 \times \$51,125) \div 365] = 68.8 \text{ days}$$

Activity or Efficiency Ratios

- Inventory Turnover Ratio: Measures the activity, or liquidity, of a firm's inventory.

$$\text{Inventory Turnover Ratio} = \text{Cost of Goods Sold} \div \text{Inventory}$$

- Average age of inventory: Average number of days' sales in inventory.

$$\text{Average Age of Inventory} = 365 \div \text{Inventory Turnover Ratio}$$

Activity or Efficiency Ratios

- The **total asset turnover** is a measure of the efficiency with which the firm uses its assets to generate sales. This ratio simply indicates how many dollars of sales a firm produces for each dollar of assets that it has invested in the business

$$\text{Total Asset Turnover} = \text{Sales} \div \text{Total Assets}$$

For Target, the total asset turnover figures for 2018 to 2020 are

$$2020: \$78,122 \div \$42,779 = 1.83$$

$$2019: \$75,356 \div \$41,290 = 1.83$$

$$2018: \$72,714 \div \$40,303 = 1.80$$

Activity or Efficiency Ratios

- Inventory Turnover Ratio: Measures the activity, or liquidity, of a firm's inventory.

$$2020: \$54,864 \div \$8,992 = 6.10$$

$$2019: \$53,299 \div \$9,497 = 5.61$$

$$2018: \$51,125 \div \$8,597 = 5.95$$

- Average age of inventory: Average number of days' sales in inventory.

$$2020: 365 \div 6.10 = 60 \text{ days}$$

$$2019: 365 \div 5.61 = 65 \text{ days}$$

$$2018: 365 \div 5.95 = 61 \text{ days}$$

Profitability Ratios

- There are many measures of profitability. As a group, these measures enable analysts to evaluate the firm's profits with respect to its sales, assets, or the owners' investment, and they are among the most closely scrutinized of all financial ratios.
- GROSS PROFIT MARGIN: The **gross profit margin** measures the percentage of each sales dollar remaining after the firm has paid for its cost of goods sold. The higher the gross profit margin is, the better.

$$\text{Gross Profit Margin} = \frac{\text{Sales} - \text{Cost of Goods Sold}}{\text{Sales}} = \frac{\text{Gross Profit}}{\text{Sales}}$$

$$2020: \frac{\$78,112 - \$54,864}{\$78,112} = 0.298 = 29.8\%$$

$$2019: \frac{\$75,356 - \$53,299}{\$75,356} = 0.293 = 29.3\%$$

$$2018: \frac{\$72,714 - \$51,125}{\$72,714} = 0.297 = 29.7\%$$

Profitability Ratios

- Net Profit Margin: The **net profit margin** measures the percentage of each sales dollar remaining after all costs and expenses, including interest, taxes, and preferred stock dividends, have been deducted. Naturally, managers and analysts view a high net profit margin in a positive light. The higher the firm's net profit margin is, the better.

$$\frac{\text{Net profit after taxes}}{\text{Net sales}}$$

$$\text{Net Profit Margin} = \text{Earnings Available for Common Stockholders} \div \text{Sales}$$

$$2020: \$3,281 \div \$78,112 = 0.042 = 4.2\%$$

$$2019: \$2,937 \div \$75,356 = 0.039 = 3.9\%$$

$$2018: \$2,914 \div \$72,714 = 0.040 = 4.0\%$$

Profitability Ratios

- EARNINGS PER SHARE (EPS): The firm's earnings per share (EPS) is generally of interest to present or prospective stockholders and management.

$$\text{Earnings per Share} = \frac{\text{Earnings Available for Common Stockholders}}{\text{Number of Shares of Common Stock Outstanding}}$$

$$2020: \$3,281 \div 511 = \$6.42$$

$$2019: \$2,937 \div 529 = \$5.55$$

$$2018: \$2,914 \div 547 = \$5.33$$

Profitability Ratios

- Return On Total Assets (ROA): The **return on total assets (ROA)**, a ratio that is sometimes called the *return on investment (ROI)*, measures the overall effectiveness of management in generating profits with its available assets.

$$\frac{\text{Net profit after taxes}}{\text{Total assets}}$$

$$\text{ROA} = \text{Earnings Available for Common Stockholders} \div \text{Total Assets}$$

$$2020: \$3,281 \div \$42,779 = 0.077 = 7.7\%$$

$$2019: \$2,937 \div \$41,290 = 0.071 = 7.1\%$$

$$2018: \$2,914 \div \$40,303 = 0.072 = 7.2\%$$

Profitability Ratios

- The **return on equity (ROE)** ratio is similar to ROA, except that it focuses on the earnings that a company generates relative to the equity invested in the firm rather than the assets invested.

$$\text{ROE} = \frac{\text{Earnings Available for Common Stockholders}}{\text{Common Stock Equity}}$$

$$\frac{\text{Net profit after taxes}}{\text{Shareholders' equity}}$$

$$2020: \$3,281 \div \$11,833 = 0.277 = 27.7\%$$

$$2019: \$2,937 \div \$11,297 = 0.260 = 26.0\%$$

$$2018: \$2,914 \div \$11,651 = 0.250 = 25.0\%$$

ROI and the Du Pont Approach

- Du Pont approach has special relevance to understanding a firm's return on investment. When we multiply the net profit margin of the firm by the total asset turnover, we obtain the return on investment, or *earning power* on total assets.
- Neither the net profit margin nor the total asset turnover ratio by itself provides an adequate measure of overall effectiveness. The net profit margin ignores the utilization of assets, and the total asset turnover ratio ignores profitability on sales.
- The return on investment ratio, or *earning power*, resolves these shortcomings. An improvement in the earning power of the firm will result if there is an increase in turnover on assets, an increase in the net profit margin, or both.

$$\begin{array}{lclcl} \text{Earning power} & = & \text{Sales profitability} & \times & \text{Asset efficiency} \\ \text{ROI} & = & \text{Net profit margin} & \times & \text{Total asset turnover} \end{array}$$

Earning power = Sales profitability × Asset efficiency

Return on investment (ROI) = $\frac{\text{Net profit after taxes}}{\text{Total assets}}$

Measures overall effectiveness in generating profits with available assets

=

Net profit margin = $\frac{\text{Net profit after taxes}}{\text{Net sales}}$

Measures profitability with respect to sales generated

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Total asset turnover = $\frac{\text{Net sales}}{\text{Total assets}}$

Measures efficiency in using assets to generate sales

$$\text{ROA} = \frac{\text{Earnings available for common stockholders}}{\text{Sales}} \times \frac{\text{Sales}}{\text{Total assets}} = \frac{\text{Earnings available for common stockholders}}{\text{Total assets}}$$

ROE and Du Pont Approach

- ROE ratio tells us the earning power on shareholders' book value investment, and is frequently used in comparing two or more firms in an industry.
- A high return on equity often reflects the firm's acceptance of strong investment opportunities and effective expense management.
- However, if the firm has chosen to employ a level of debt that is high by industry standards, a high ROE might simply be the result of assuming excessive financial risk.
- The second step in the DuPont system employs the **modified DuPont formula**. This formula relates the firm's return on total assets (ROA) to its return on equity (ROE). The key ratio linking ROA to ROE is the **financial leverage multiplier (FLM)**, which is the ratio of total assets to common stock equity

$$\text{Financial Leverage Multiplier (FLM)} = \text{Total Assets} \div \text{Common Stock Equity}$$

ROE and Du Pont Approach

- Next, it's easy to see that the product of ROA and FLM equals the firm's ROE.

$$\text{ROE} = \text{ROA} \times \text{FLM}$$

Substituting the appropriate formulas for ROA and FLM into the equation and simplifying results in the formula for ROE given earlier:

$$\text{ROE} = \frac{\text{Earnings available for common stockholders}}{\text{Total assets}} \times \frac{\text{Total assets}}{\text{Common stock equity}} = \frac{\text{Earnings available for common stockholders}}{\text{Common stock equity}}$$

- ROE Du Pont approach – that is, break this return measure into its components.

$$\begin{aligned} \frac{\text{Net profit after taxes}}{\text{Shareholders' equity}} &= \frac{\text{Net profit after taxes}}{\text{Net sales}} \times \frac{\text{Net sales}}{\text{Total assets}} \times \frac{\text{Total assets}}{\text{Shareholders' equity}} \\ \text{ROE} &= \text{Net profit margin} \times \text{Total asset turnover} \times \text{Equity multiplier} \end{aligned}$$

Market Ratios

- **Market ratios** relate the firm's market value, as measured by its current share price, to certain accounting values.
- These ratios give insight into how investors in the marketplace believe the firm is doing in terms of risk and return.
- An interesting aspect of these ratios is that they combine backward-looking and forwardlooking perspectives.
- A firm's stock price is intrinsically forward looking because what investors are willing to pay for a stock is based much more on how they think a company will perform in the future than on how it has performed in the past.
- Accounting values, in contrast, have an inherently historical perspective.

Market Ratios

- PRICE/EARNINGS (P/E) RATIO: The **price/earnings (P/E) ratio** is one of the most widely quoted financial ratios in the investment community.
- The P/E ratio reveals what investors are willing to pay for each dollar of a firm's earnings. A variety of factors can influence whether a firm's P/E ratio is high or low, but one of the primary determinants of the P/E ratio is the rate of growth that investors believe a firm will achieve.

$$\text{P/E Ratio} = \text{Market Price per Share of Common Stock} \div \text{Earnings per Share}$$

- Investors will pay a higher price for the shares of a firm they expect to grow rapidly, so fast-growing firms tend to have higher P/E ratios than more stable companies.
- Another factor that influences this ratio is risk. Holding everything else constant, investors will pay a higher price for a firm they believe is less risky. Therefore, the P/E ratio may be high if a firm's growth prospects are good, if its risk is low, or (as we will soon see) for other reasons.

Market Ratios

- PRICE/EARNINGS (P/E) RATIO:

$$\text{P/E Ratio} = \text{Market Price per Share of Common Stock} \div \text{Earnings per Share}$$

$$2020: \$110.74 \div \$6.42 = 17.25$$

$$2019: \$73.00 \div \$5.55 = 13.15$$

$$2018: \$75.00 \div \$5.33 = 14.07$$

Market Ratios

- MARKET/BOOK (M/B) RATIO: The **market/book (M/B) ratio** provides an assessment of how investors view the firm by comparing the market price of the firm's common stock to its book value (i.e., the value shown on the balance sheet). For most companies, the M/B ratio will be greater than one, and often it is much greater than one.

$$\text{Market/Book (M/B) Ratio} = \frac{\text{Market Price per Share of Common Stock}}{\text{Book Value per Share of Common Stock}}$$

$$2020: \$110.74 \div \$23.16 = 4.78$$

$$2019: \$73.00 \div \$21.36 = 3.42$$

$$2018: \$75.00 \div \$21.30 = 3.52$$

$$\text{Book Value per Share of Common Stock} = \frac{\text{Common Stock Equity}}{\text{Number of Shares of Common Stock Outstanding}}$$

Common-Size Analysis

- An analysis of *percentage* financial statements where all balance sheet items are divided by *total assets* and all income statement items are divided by *net sales or revenues*.

ASSETS	REGULAR (IN THOUSANDS)			COMMON-SIZE (%)		
	20X0	20X1	20X2	20X0	20X1	20X2
Cash	\$ 2,507	\$ 11,310	\$ 19,648	1.0	3.8	5.1
Accounts receivable	70,360	85,147	118,415	29.3	28.9	30.9
Inventory	77,380	91,378	118,563	32.2	31.0	31.0
Other current assets	6,316	6,082	5,891	2.6	2.1	1.5
Current assets	\$156,563	\$193,917	\$262,517	65.1	65.8	68.5
Fixed assets, net	79,187	94,652	115,461	32.9	32.2	30.1
Other long-term assets	4,695	5,899	5,491	2.0	2.0	1.4
Total assets	<u>\$240,445</u>	<u>\$294,468</u>	<u>\$383,469</u>	<u>100.0</u>	<u>100.0</u>	<u>100.0</u>
LIABILITIES AND SHAREHOLDERS' EQUITY						
Accounts payable	\$ 35,661	\$ 37,460	\$ 62,725	14.8	12.7	16.4
Notes payable	20,501	14,680	17,298	8.5	5.0	4.5
Other current liabilities	11,054	8,132	15,741	4.6	2.8	4.1
Current liabilities	\$ 67,216	\$ 60,272	\$ 95,764	27.9	20.5	25.0
Long-term debt	888	1,276	4,005	0.4	0.4	1.0
Total liabilities	\$ 68,104	\$ 61,548	\$ 99,769	28.3	20.9	26.0
Common stock	12,650	20,750	24,150	5.3	7.0	6.3
Additional paid-in capital	37,950	70,350	87,730	15.8	23.9	22.9
Retained earnings	121,741	141,820	171,820	50.6	48.2	44.8
Total shareholders' equity	<u>\$172,341</u>	<u>\$232,920</u>	<u>\$283,700</u>	<u>71.7</u>	<u>79.1</u>	<u>74.0</u>
Total liabilities and shareholders' equity	<u>\$240,445</u>	<u>\$294,468</u>	<u>\$383,469</u>	<u>100.0</u>	<u>100.0</u>	<u>100.0</u>

Common-Size Analysis

	REGULAR (IN THOUSANDS)			COMMON-SIZE (%)		
	20X0	20X1	20X2	20X0	20X1	20X2
Net sales	\$323,780	\$375,088	\$479,077	100.0	100.0	100.0
Cost of goods sold	<u>148,127</u>	<u>184,507</u>	<u>223,690</u>	<u>45.8</u>	<u>49.2</u>	<u>46.7</u>
Gross profit	\$175,653	\$190,581	\$255,387	54.2	50.8	53.3
Selling, general, and administrative expenses	131,809	140,913	180,610	40.7	37.6	37.7
Depreciation	7,700	9,595	11,257	2.4	2.5	2.3
Interest expense	<u>1,711</u>	<u>1,356</u>	<u>1,704</u>	<u>0.5</u>	<u>0.4</u>	<u>0.4</u>
Earnings before taxes	\$ 34,433	\$ 38,717	\$ 61,816	10.6	10.3	12.9
Taxes	<u>12,740</u>	<u>14,712</u>	<u>23,490</u>	<u>3.9</u>	<u>3.9</u>	<u>4.9</u>
Earning after taxes	<u>\$ 21,693</u>	<u>\$ 24,005</u>	<u>\$ 38,326</u>	<u>6.7</u>	<u>6.4</u>	<u>8.0</u>

Index Analyses

- An analysis of *percentage* financial statements where all balance sheet or income statement figures for a base year equal 100.0 (percent) and subsequent financial statement items are expressed as percentages of their values in the base year.

	REGULAR (IN THOUSANDS)			INDEXED (%)		
	20X0	20X1	20X2	20X0	20X1	20X2
Net sales	\$323,780	\$375,088	\$479,077	100.0	115.8	148.0
Cost of goods sold	<u>148,127</u>	<u>184,507</u>	<u>223,690</u>	<u>100.0</u>	<u>124.6</u>	<u>151.0</u>
Gross profit	\$175,653	\$190,581	\$255,387	100.0	108.5	145.4
Selling, general, and administrative expenses	131,809	140,913	180,610	100.0	106.9	137.0
Depreciation	7,700	9,595	11,257	100.0	124.6	146.2
Interest expense	<u>1,711</u>	<u>1,356</u>	<u>1,704</u>	<u>100.0</u>	<u>79.3</u>	<u>99.6</u>
Earnings before taxes	\$ 34,433	\$ 38,717	\$ 61,816	100.0	112.4	179.5
Taxes	<u>12,740</u>	<u>14,712</u>	<u>23,490</u>	<u>100.0</u>	<u>115.5</u>	<u>184.4</u>
Earnings after taxes	<u>\$ 21,693</u>	<u>\$ 24,005</u>	<u>\$ 38,326</u>	<u>100.0</u>	<u>110.7</u>	<u>176.7</u>

Index Analyses

ASSETS	REGULAR (IN THOUSANDS)			INDEXED (%)		
	20X0	20X1	20X2	20X0	20X1	20X2
Cash	\$ 2,507	\$ 11,310	\$ 19,648	100.0	451.1	783.7
Accounts receivable	70,360	85,147	118,415	100.0	121.0	168.3
Inventory	77,380	91,378	118,563	100.0	118.1	153.2
Other current assets	6,316	6,082	5,891	100.0	96.3	93.3
Current assets	<u>\$156,563</u>	<u>\$193,917</u>	<u>\$262,517</u>	<u>100.0</u>	<u>123.9</u>	<u>167.7</u>
Fixed assets, net	79,187	94,652	115,461	100.0	119.5	145.8
Other long-term assets	4,695	5,899	5,491	100.0	125.6	117.0
Total assets	<u><u>\$240,445</u></u>	<u><u>\$294,468</u></u>	<u><u>\$383,469</u></u>	<u><u>100.0</u></u>	<u><u>122.5</u></u>	<u><u>159.5</u></u>
LIABILITIES AND SHAREHOLDERS' EQUITY						
Accounts payable	\$ 35,661	\$ 37,460	\$ 62,725	100.0	105.0	175.9
Notes payable	20,501	14,680	17,298	100.0	71.6	84.4
Other current liabilities	11,054	8,132	15,741	100.0	73.6	142.4
Current liabilities	<u>\$ 67,216</u>	<u>\$ 60,272</u>	<u>\$ 95,764</u>	<u>100.0</u>	<u>89.7</u>	<u>142.5</u>
Long-term debt	888	1,276	4,005	100.0	143.7	451.0
Total liabilities	<u>\$ 68,104</u>	<u>\$ 61,548</u>	<u>\$ 99,769</u>	<u>100.0</u>	<u>90.4</u>	<u>146.5</u>
Common stock	12,650	20,750	24,150	100.0	164.0	190.9
Additional paid-in capital	37,950	70,350	87,730	100.0	185.4	231.2
Retained earnings	<u>121,741</u>	<u>141,820</u>	<u>171,820</u>	<u>100.0</u>	<u>116.5</u>	<u>141.1</u>
Total shareholders' equity	<u>\$172,341</u>	<u>\$232,920</u>	<u>\$283,700</u>	<u>100.0</u>	<u>135.2</u>	<u>164.6</u>
Total liabilities and shareholders' equity	<u><u>\$240,445</u></u>	<u><u>\$294,468</u></u>	<u><u>\$383,469</u></u>	<u><u>100.0</u></u>	<u><u>122.5</u></u>	<u><u>159.5</u></u>